Diabetes Care Organization, Process, and Patient Outcomes: Effects of a Diabetes Control Program

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Effects of a diabetes control program were evaluated in 17 primary health centers after 18 months' participation (17 nonparticipating centers served as controls). Studied were the effects of the program on the organization of diabetes care, on the process of care in terms of services delivered to patients, and on the outcomes of care as measured by improved self-care, dietary knowledge, and HbA_{1c} values. While findings showed that patients from the intervention centers received a higher quality of service and monitored their blood glucose levels more often than did patients from the control centers, they did not demonstrate significantly better metabolic control. The reasons for this, and ways to improve the program are discussed.

In recent years, diabetes care in Sweden, as in many other countries, has been a shared responsibility between hospital and primary caregivers.[1] A Diabetes Control Program was initiated in 1979 that incorporated the latest findings from biomedical research in diabetology, and standards for organization of care.[2] In the Stockholm area (1.6 million inhabitants), a special Diabetes Education and Training Unit (DETU) was established to implement the Diabetes Control Program in primary care. Between 1980 and 1984, this was accomplished through continuing medical education (CME) courses. An evaluation identified important obstacles to the implementation of organizational changes, including poor knowledge in diabetology, poor cooperation among staff members, lack of adequate resources, and lack of guidelines for diabetic care.[3,4] These results corroborated similar follow-ups of traditional CME courses. Thus, even when adequate knowledge was transferred, this form of education seldom affected physician's behavior or the organization of care at the local level.[5,6]

To overcome these obstacles, the DETU developed a new implementation strategy that combined CME and organizational development (OD) measures. This strategy was evaluated on two levels: effects on the implementation of organizational changes in health centers; and effects on staff competence in diabetes-specific areas, including resources available for high-quality diabetes care, involvement with diabetes care, and cooperation among staff members.[8] The results showed that the combined CME and OD approach had a strongly positive influence on all of the above parameters.[7,8]

Because the DETU interventions were not intended to directly influence patient outcomes, we designed the present study to evaluate the DETU's effect on the organization and process of diabetic care on patients' health and well-being (Fig 1). To do this, we needed answers to the following questions:

1. Does participation in the DETU program and nurse involvement, resources, and competence affect the quality of care?

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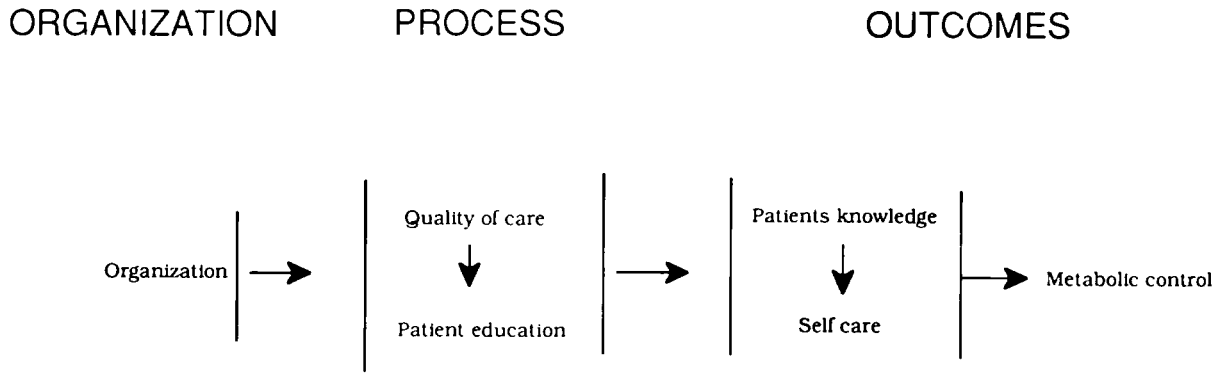

Fig 1. *Model of diabetes care organization, process, and outcomes.*

2. Do care routines in health centers, or quality factors such as continuity of, and access to, care contribute to the amount of patient education?
3. Does patient education contribute to patients' knowledge or self-care activities?
4. Do patients' knowledge or self-care habits improve metabolic control?

Methods

Study Design Diabetes care was evaluated at 34 primary health care centers (PHCC) that either had been exposed to the combined CME-OD program for 18 months (17 centers, intervention group), or were about to enter the program (17 centers, control group). The centers had been randomly assigned to the two groups. The intervention program and randomization process have been described in detail elsewhere.[7-9]

Data Patients with diabetes who had visited the PHCC during the 12 months prior to the evaluation made up the study population. Data on height, HbA_{1c} concentration, and results of urine testing for proteinuria were obtained from the patients' case records. These data were used as indicators of follow-up routines and control of late complications, and were considered part of the quality of services.

From each center's register of diabetic patients a 20% random sample was selected (secondary sampling).[10] These patients were asked to complete a questionnaire and had a blood sample taken by DETU staff for HbA_{1c} measurement. The questionnaire was designed to yield information on the following factors:

1. The quality of services as perceived by patients in terms of
- *access to care*, measured by the patients' satisfaction with the number of visits to either a physician or a nurse per year (both yes/no);
- *the continuity of care*, measured by the frequency of seeing the same physician or nurse (always, often/sometimes, never);
- the date of the most recent *eye examination* (during last year or earlier); and
- *a global rating of quality of diabetes care* encountered (very good, good/not so good, bad).

2. Information on diabetes-specific issues, including
- information delivered by health care providers at the health center on diet, foot care, self-monitoring of blood glucose (SMBG) levels, and late complications of diabetes (in general), kidney, foot, and eye problems; coronary heart disease (complications specified); and hypoglycemia (all yes/no);
- main source of information at the health center—physician, nurse, dietitian or chiropodist (all yes/no);
- whether the patient had actively obtained information on diabetes from sources outside the health center (yes/no);
- whether any member of the patient's family had received any information on diabetes at the health center (yes/no).

3. Knowledge as measured by the patients' responses to eight multiple-choice questions designed to test knowledge about diet and nutrition (maximum 18 points). Six of the eight questions were used by Karlander.[11]

4. Self-care in terms of SMBG or urine testing, dietary changes after onset, and self-examination of feet (all yes/no).

All physicians and nurses in the 34 PHCCs were asked to fill in a questionnaire to assess involvement in diabetic care, their perception of the diabetes-specific resources available to them in the health center, knowledge (responses to three case history tests), and self-rated competence. Details of these measures have been reported elsewhere.[8]

In this study we used:

1. The proportion of nurses involved in diabetes care (four or more of seven specified tasks taken on regularly)[8]
2. Mean number of resources available to nurses involved (range, 0 through 9)[8]
3. Mean number of items regarding self-rated competence positively responded to ("feel confident to handle . . . ") by nurses involved in diabetic care (range, 0 through 12)[8,12]
4. Mean number of case history tests answered correctly by nurses involved in diabetic care (range, 0 through 3)[8,13]

The following indexes were constructed from unweighted sums of variables within different factors in the multiple regression analysis (see below):

- "Routines," as assessed by means of an audit of patients' case records—the portion of the records that included the patient's height, HbA_{1c} values, and urine test results for proteinuria (range, 49 through 264)
- "Quality of service," patients' responses to the questions about the availability, continuity, quality of care, and eye examination (sum of positive answers; range, 0 through 6)

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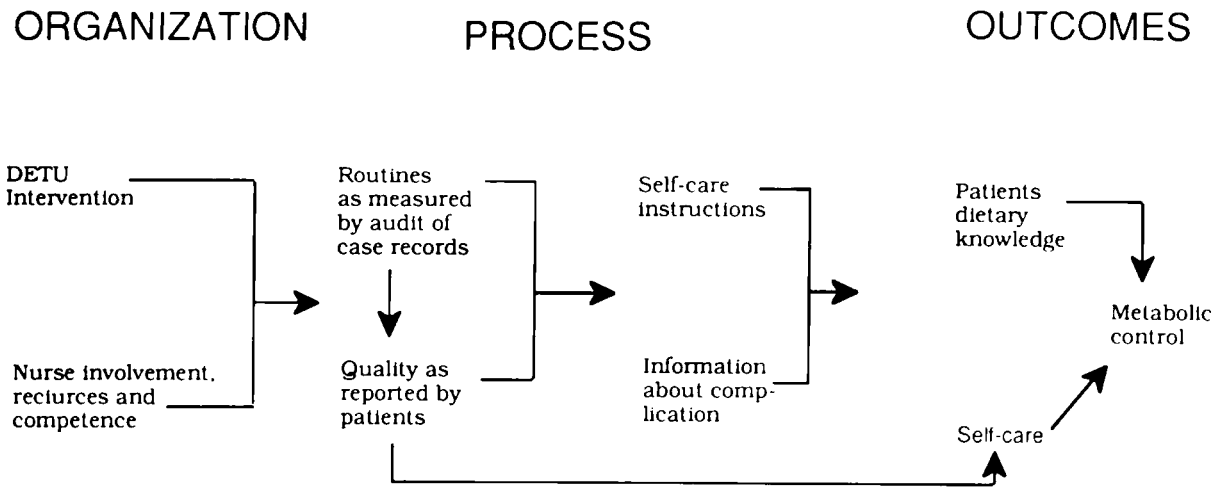

Fig 2. *Associations tested in multiple regression analysis.*

- "Self-care instructions," the sum of positive answers to the items on information received in the health center about diet, SMBG, foot care, and late complications in general (range, 0 through 4)
- "Information on diabetic complications," the sum of positive answers to the specified items on information received from the health center about late complications: kidney, eye, and foot problems, and coronary heart disease (range, 0 through 4)
- "Dietary knowledge," sum of correct answers to eight questions on diet (range, 0 through 18)
- "Self-care," sum of the positive answers to items on SMBG or urine testing, dietary changes, and foot inspection (range, 0 through 3)

Multiple correlations were tested as outlined in Fig 2:

- The contributions of intervention and percentage of nurses involved, involved nurses' access to resources, self-rated competence, and results on the case history test to the variance in the two quality measures (routines and quality of service)
- The contributions of the two quality measures to the variance in the two measures of patient education (self-care instructions and information about complications)
- The contribution of information (self-care instructions only) to patients' knowledge
- The contribution of routines, quality of service, self-care instructions, information on diabetic complications, and knowledge to variance in self-care and metabolic control

The patients' age, duration of diabetes, type of treatment, and formal education have been controlled for in all analyses.

Analysis Differences between the intervention and control groups were studied with χ^2 and Student's *t*-tests. Since it was the health centers that were randomized, not the individual patients, multivariate analysis was performed by means of multiple regression.[14]

Results

A total of 4,492 patients with diabetes had visited the 34 PHCCs in the 12 months prior to the evaluation. Patients from the two groups of health centers (17 intervention and 17 control centers) were similar with regard to *treatment*: diet only (36%, 39%), oral medication (47%, 45%), insulin treatment (17%, 16%); *male/female ratio*: (49%/51%, 44%/56%); mean *age* (M ± SD): (64.3 ± 12.1, 63.9 ± 11.5); and mean *duration of disease* (M ± SD): (6.4 ± 7.0, 5.5 ± 5.9). Neither were there any differences between the intervention and control groups, respectively, with regard to formal *education*: high school or above (15%, 12%); or *work status*: old-age pensioner (61%, 58%), working full or part time (26%, 29%), and prematurely retired (14%, 15%).

The 20% secondary sample comprised 806 patients, of whom 566 (70%) answered the questionnaire and had their HbA_{1c} value measured. Of the 240 patients in the secondary sample who did not participate, 69 could not be reached and 171 refused to take part. Seventy-three percent (n = 317) of subjects selected in the intervention group participated in the study, as did 69% (n = 249) of those from the control group.

Participating patients in the secondary sample did not differ from the total sample with respect to age, male/female ratio, duration of diabetes, treatment, or latest blood glucose value noted in the case records.

Comparisons Between Intervention and Control Groups of Centers Significant differences between the two groups of centers (intervention versus control) were found in the following areas: patient's height noted in case notes (73%, 50%, $P<.01$), HbA_{1c} value measured (27%, 8%, $P<.001$), and eye examination performed during previous year (40%, 28%, $P<.01$). There were no differences found between the two groups regarding access to care or its continuity. The majority of patients rated *access to care* as fairly high—90% of patients from the intervention group and 89% from the control group were satisfied with number of visits to the physician; and 88% and 79%, respectively, were satisfied with the number of visits to the nurse. Physician *continuity* was also rated positively, with 77% of patients from the intervention group and 75% from the control group reporting that they saw the same doctor at every or almost every visit. Nurse continuity was rated positively somewhat less often, with 60% and 56% of patients, respectively, reporting having seen the same nurse at every or almost every visit.

No significant differences were observed with regard to *patient education*: 76% of the patients in both groups (interven-

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tion versus control group) reported receiving dietary information; 67% and 62%, respectively, received information on SMBG; 50% and 42% received information on preventive foot care; 18% and 19%, on late complications in general; 38% and 29%, on foot complications; 18% in both groups, on kidney problems; 39% and 36%, on eye problems; 25% and 24%, on coronary heart disease; and 20% and 25%, on hypoglycemia. Neither did the two groups of patients (intervention versus control) differ on whom they rated as the most important provider of information. The physician was the *main source of information* for 53% and 55% of patients, respectively; the nurse for 37% and 40%; the dietitian for 29% and 28%; and the chiropodist for 4% and 8%. Fifty-eight percent of patients in the intervention centers and 53% of patients in the control centers reported having *obtained information from outside the health care center*, while 20% in the intervention group and 17% in the control group reported that a *family member* received information about diabetes from the health center.

No difference was found in *dietary knowledge* between patients from intervention and control groups of centers. On average, patients from intervention centers marked 10.3 ± 3.5 ($M \pm SD$) of 18 subitems correctly, while patients from control centers marked 10.5 ± 3.3 subitems correctly.

Patients from the intervention group of health centers reported significantly more often than did patients from the control group that they used *SMBG* (37% and 23%, respectively, $P<.001$). No differences were found between intervention and control groups on the other three items—84% and 86%, respectively, reported *dietary changes* after onset, 65% in both groups *examined their feet*, and 42% and 50% of patients treated with insulin did *modify their insulin dosages*.

The two groups of patients also had a similar degree of metabolic control: the mean HbA_{1c} *value* was 8.1 ± 1.8 and 7.8 ± 1.6 ($M \pm SD$) in the intervention and control groups, respectively.

Multivariate Analysis Results are shown in Tables 1 through 5 and summarized in Fig 3. Significantly contributing to the variance in *routines* (patient's height noted in case records, HbA_{1c} level measured, and urine tested for proteinuria) were: availability of resources, self-rated competence, and results on case history test among nurses involved in diabetes care; intervention (health center involved in the DETU program); age of patient (negative contribution: patients from centers with high values on the index were younger) (Model $R^2 = 0.50$, Table 1).

The patients' perception of the *quality of services* showed a different pattern. Here intervention had no independent effect. There was a negative association between the number of nurses looking after patients and the patients' assessment of the quality of care. Thus, the more nurses involved, the lower the patients' rating of quality in terms of continuity of care, access to care, and so forth ($\beta = -0.19$, $P = .048$).* Apart from the number of nurses involved, only the nurses' self-rated competence ($\beta = 0.08$, $P = .04$) and patients' duration of diabetes ($\beta = 0.03$, $P = .01$) contributed significantly to quality of service (Model $R^2 = 0.07$, data not shown).

To *self-care instructions* (on diet, SMBG, preventive foot care, and late complications in general), quality of service was the most important contributing factor. Other factors that contributed significantly were routines, whether or not a family member had received information about diabetes at the health center, whether or not a nurse or physician had been

Table 1. Multiple Regression Analysis With Routines (range 49 through 264) as Dependent Variable.

Predictor	β	P	Model R^2
Intercept	103.7		
Intervention*	−34.8	<.001	
Case history test †	10.0	=.001	
Resources ‡	8.0	<.001	
Self-rated competence §	3.3	=.007	
Age of patient	−0.4	=.040	0.50

Note: Competing factors in the multivariate analysis were: intervention, nurses involved in diabetic care, resources available, self-rated competence and knowledge as measured by the case history test among nurses involved, and patients' age, duration of diabetes, treatment, and educational level. Only factors significant at the 5% level are presented.

* Health center involved in DETU program or not, yes = 0/no = 1.
† Average number of correct answers from nurses involved with diabetic care in health center, range 0 through 3.
‡ Average number of resources available to nurses involved in diabetic care, range 0 through 9.
§ Average number of items positively responded to by nurses involved in diabetic care, range 0 through 12.

Table 2. Multiple Regression Analysis With Self-Care Instructions (range 0 through 4) as the Dependent Variable

Predictor	β	P	Model R^2
Intercept	1.2		
Nurse as main source of information*	0.5	=.004	
Physician as main source of information*	0.4	=.018	
Family member informed*	0.4	=.001	
Service quality †	0.3	<.001	
Age of patient	−0.02	<.001	
Routines ‡	0.004	=.028	0.26

Note: Competing factors in the analysis were: routines, quality of services, main source of information, information obtained from outside health care center, whether or not a family member had received information about diabetes at the health center, patients' age, duration of diabetes, treatment, and educational level. Only factors significant at the 5% level are presented.

*Yes = 1; no = 0.
†Range, 0 through 6.
‡Range, 49 through 264.

* Range 0 through 4, nurse involvement was coded as follows: <25% of nurses involved = 0; 25% to 50% of nurses involved = 1; 51% to 75% of nurses involved = 2; and >75% of nurses involved = 3.

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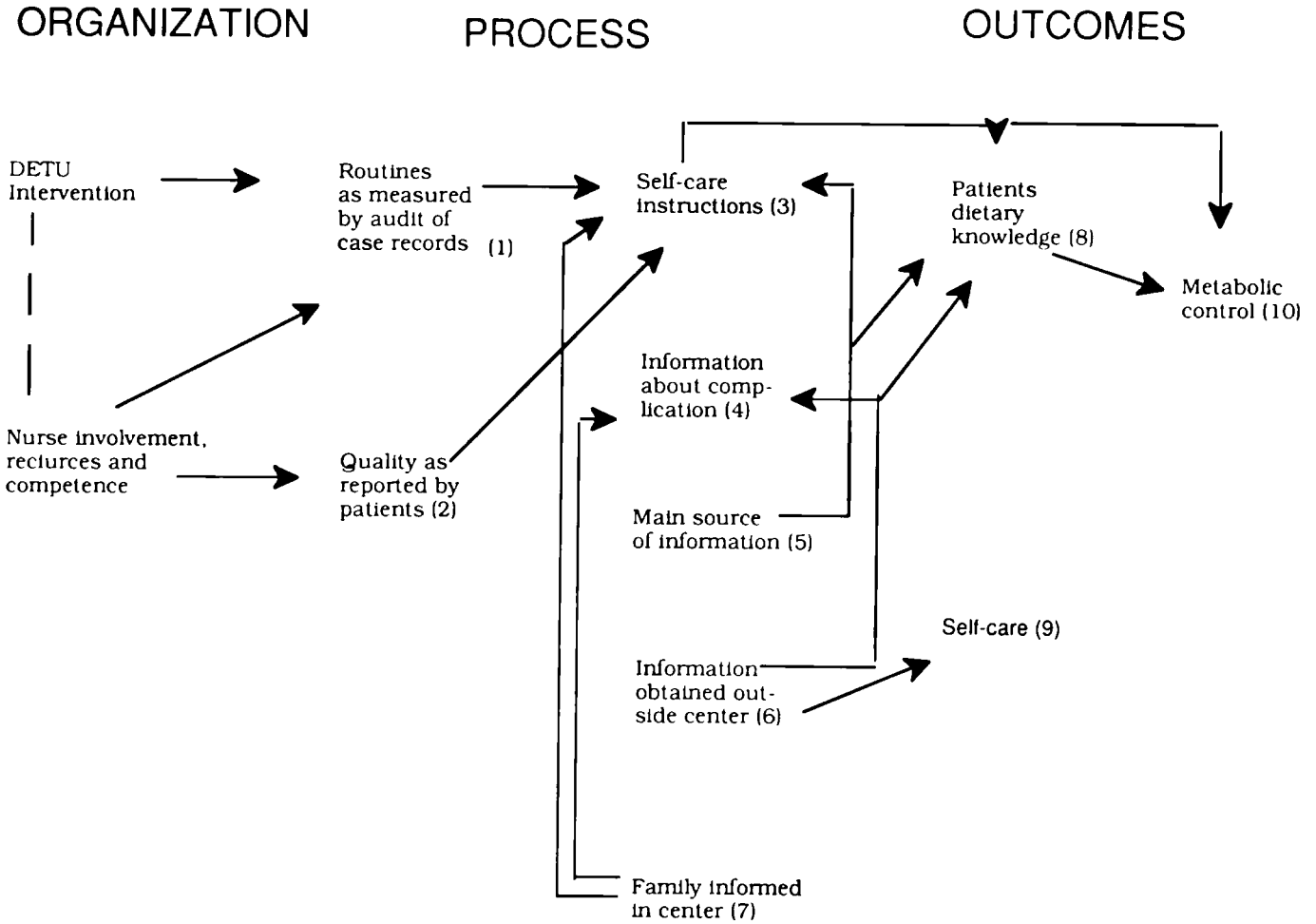

Fig 3. *Summary of results: solid line indicates significant correlation in multiple regression analysis. Numbers in parentheses indicate the following: (1) patient's height noted in case record, HbA_{1c} concentration measured, and urine tested for proteinuria; (2) patient's satisfaction with access to care by physician/nurse, continuity of care, global rating of quality of care, and eyes examined during previous 12-month period; (3) patient informed in health center about diet, preventive foot care, SMBG, and late complications; (4) patient informed in health care center about eye, kidney, and foot problems and coronary heart disease; (5) whether main source of information at center was a nurse, physician, dietitian, or chiropodist; (6) whether patient obtained any information/education from sources other than health care personnel at center; (7) whether member of patient's family received any information/education; (8) score on knowledge test; (9) attention to self-care in terms of SMBG or urine testing, dietary changes after onset, and regular foot inspection; and (10) metabolic control as reflected in HbA_{1c} value. In addition, it should be noted that patient's age was significantly associated with (1), (3), and (8); duration of diabetes with (2); and treatment with (9) and (10).*

the main source of the information received in health center, and age of patient (negative contribution: older patients reported less information received) (Model $R^2 = 0.26$, Table 2).

Information from outside the health center ($\beta = 0.69$, $P = .01$), and whether or not a family member had received information about diabetes ($\beta = 0.81$, $P = .02$) contributed significantly to *information about complications* (complications specified) (Model $R^2 = 0.11$, data not shown).

Factors that significantly contributed to the variance in patients' *dietary knowledge* were self-care instructions, whether or not a dietitian was the main source of information, whether or not information had been obtained from outside the center, and the age of the patient (negative contribution: older patients less knowledgeable) (Model $R^2 = 0.20$, Table 3).

As expected, patients who were more knowledgeable about diet were also more actively involved with *self-care*, as were patients who had obtained information from outside the health center and had less complicated treatment (Model $R^2 = 0.24$, Table 4). Finally, patients with less complicated treatment who scored low on self-care instructions and high on

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Table 3. Multiple Regression Analysis With Patients' Dietary Knowledge (range 0 through 18) as the Dependent Variable

Predictor	β	P	Model R^2
Intercept	6.3		
Information from outside health center*	2.1	<.001	
Dietitian as main source of information*	1.5	=.007	
Self-care instructions †	0.7	<.001	
Age	−0.1	=.015	0.20

Note: Competing factors in the analysis were: self-care instructions, information obtained from outside health care center, whether or not a family member had received information about diabetes at the health center, main source of information (physician, nurse, dietitian, chiropodist), patients' age, duration of diabetes, treatment, and educational level. Only factors significant at the 5% level are presented.

*Yes = 1; no = 0.
†Range, 0 through 4.

Table 4. Multiple Regression Analysis With Self-Care (range 0 through 3) as Dependent Variable

Predictor	β	P	Model R^2
Intercept	0.9		
Treatment*	0.3	<.001	
Information from outside health center †	0.3	=.015	
Knowledge ‡	0.1	<.001	0.24

Note: Competing factors in the analysis were: self-care instructions, information from outside the health center, whether or not a family member had received information about diabetes at the health center, main source of information (physician, nurse, dietitian, chiropodist), patients' knowledge, age, duration of diabetes, treatment, and educational level. Only factors significant at the 5% level are presented.

*Insulin = 2, oral medication = 1, diet alone = 0.
†Yes = 1; no = 0.
‡Range, 0 through 18.

Table 5. Multiple Regression Analysis With HbA_{1c} Level as the Dependent Variable

Predictor	β	P	Model R^2
Intercept	75.6		
Treatment*	6.2	=.008	
Self-care instructions †	2.9	=.029	
Knowledge ‡	−1.3	=.001	0.19

Note: Competing factors in the analysis were: routines, quality of services, information about complications, information from outside the health care center, whether or not a family member had received information about diabetes at the health center, patients' knowledge, self-care, age, duration of diabetes, treatment, and educational level. Only factors significant at the 5% level are presented.

*Insulin = 2, oral medication = 1, diet alone = 0.
†Range, 0 through 4.
‡Range, 0 through 18.

the dietary-knowledge test, had the best metabolic control as measured by HbA_{1c} level (Model R^2 = 0.19, Table 5).

Discussion

The DETU program was found to have had an effect on the routines for diabetes care, nurse involvement, the resources available, and staff self-rated competence in the centers. Diabetes care routines and quality of service, as reported by the patients, were important for the patients' education, which in turn was related to their dietary knowledge, practice of self-care, and their level of metabolic control. More knowledge was also associated with a lower HbA_{1c} value.

The findings of this study fit with the model on the relations between program input and patient outcomes (Fig 1). However, our conclusions are the result of many correlations, and the problem of mass-significance should be mentioned. Thus, out of 70 comparisons, three to four significant findings could have been generated by chance (5% risk level).

Our evaluation covered a wide range of management activities both in the clinical setting and at the patient level; however, the parameters used should be viewed more as indicators of important aspects of care rather than as efforts to cover the complexity of the disease and its treatment.

Both self-care habits and metabolic control were found to be determined to some extent by patients' dietary knowledge. However, the most important contributor to the degree of metabolic control was the type of treatment (β = 6.2). Thus, patients treated by diet alone had lower HbA_{1c} values than did patients treated with oral medication. The latter group, in turn, was better controlled than were patients treated with insulin. The DETU intervention had no effect on treatment. This finding could indicate that patients were overtreated or undertreated with insulin. Alternatively, many of the patients treated with diet alone still had some residual insulin production that may have been important in controlling their diabetes.[15]

Patients who had received the most information about foods in the health center were also more knowledgeable about the effects of diet on diabetes. An equally important contributing factor was whether the patient had obtained information from sources outside the health center, which also contributed significantly to improved self-care activity. This finding shows the importance of the patient's attitude to information about diabetes[16] and "information-seeking skills,"[17] as well as the use of the student's questions and perception of the subject as the basis for education.[18,19]

The proportion of nurses involved in diabetic care correlated negatively with patients' perception of the quality of services as measured by the quality-of-service index. Involving more nurses in the intervention group may have affected such factors as continuity and access to care in a negative way, at least temporarily.

One explanation for the weak correlation between the organization of care and quality of service and the patients' degree of metabolic control, as assessed by HbA_{1c} values, may be that our evaluation was made too soon after the intervention. The patients had not been exposed to the effects of organizational changes and staff involvement, or improved competence, for any length of time. In Florida, a similar

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training of primary care staff in handling diabetes and reorganizing care was evaluated. These researchers only reported on organizational changes occurring during the 2 years following the program because it was found to be too early to measure any effects on patient outcomes.[20]

Another explanation for the limited results of our intervention program on patient outcomes could be that the linear model used (Figs 1 through 3) is too narrow, both regarding what is hypothesized as influencing quality of diabetes care and what is measured as patient outcomes.[16,21-24]

Another question of importance is whether evaluation of care or quality assurance programs should be the responsibility of external agencies like the DETU or should be systematically and regularly performed by the health center staff itself.[25,26] The value of internal performance reviews at regular intervals has also been documented[20,25,26]: "Models for quality improvement emphasize internal, organization-based, professionally led efforts to improve many small processes of care in a ceaseless cycle of examination and change."[25]

Conclusions

Results from this study show that (1) the DETU intervention and nurse involvement, resources, and competence had an effect on the quality of diabetes care; (2) that care routines in health centers and patients' perceptions of factors such as continuity of and access to care contributed to the amount of patient education; and (3) that patient education contributed to patients' dietary knowledge and, to some extent, to their metabolic control. However, correlations between organizational and process factors and patient outcomes were weak. Despite strong effects of the program on organization and process parameters, these effects were more or less lost as assessed by the last outcome measure, the patients' metabolic control. Future efforts should focus more attention on the problem of patient education in the CME program, as well as expand the model for its evaluation. Furthermore, new strategies for encouraging general practitioners and others in the health centers to institute some kind of regular self-examination or performance review has to be developed.

References

1. Luft R. Filling the gaps. Presentation at WHO Conference on Diabetes Care as a Model for Primary Health Care. Karolinska Institute, Stockholm, June 9-11, 1986.

2. Luft R. Diabetes. Underlag till vårdprogram för primärvården [Diabetes. Basis for control program in the primary care]. Malmö: Utbildningsproduktion AB, 1978.

3. Rosenqvist U, Carlson A, Luft R. Evaluation of a comprehensive program for diabetes care at primary health-care level. Diabetes Care 1988;11:269-74.

4. Carlson A, Rosenqvist U. Diabetes control program implementation: on the importance of staff involvement. Scand J Prim Health Care 1988;(suppl 1):105-12.

5. Mazze R, Deeb L, Palumbo PJ. Altering physicians' practice patterns—a nationwide educational experiment: evaluation of the clinical education program of the American Diabetes Association. Diabetes Care 1986;9:420-5.

6. Knopf RF, Kittel PR, Funnell MM, Wolf FM. Development and evaluation of diabetes continuing education courses for health professionals: a synthesis of eight years of experience. Diabetes Educ 1988;14:136-41.

7. Carlson A, Rosenqvist U. Locally developed plans for quality diabetes care. Worker and consumer participation in the public health care system. Health Educ Res 1990; 5(1): 41-51.

8. Carlson A, Rosenqvist U. Reducing the obstacles to the provision of quality diabetes care. Psychology and Health (in press).

9. Rosenthal MM, Carlson A, Rosenqvist U. Beyond CME: diabetes education field-interactive strategies from Sweden. Diabetes Educ 1988;14:212-17.

10. Cochran NG. Sampling techniques. 3rd ed. New York: John Wiley & Sons, 1977.

11. Karlander S-G. On the routine management of middle-aged diabetic patients [Dissertation]. Stockholm, Sweden: Karolinska Institute, 1985

12. Mellor JG, Samanta A, Blandford RL, Burden AC. Questionnaire survey of diabetic care in general practice in Leicestershire. Health Trends 1985;17:61-63.

13. Griffiths K, McDevitt DG, Andre M. et al. Therapeutic traditions in Northern Ireland, Norway and Sweden: I. Diabetes. WHO Drug Utilization Research Group (DRUG). Eur J Clin Pharmacol 1986;30:513-19.

14. Kleinbaum DG, Kupper LL. Applied regression analysis and other multivariable methods. Boston: Duxbury Press, 1978.

15. Sjöberg S, Carlson A, Rosenqvist U, Östman J. Health attitudes, self-monitoring of blood glucose, metabolic control and residual insulin secretion in type 1 diabetic patients. Diabetic Med 1988;5:449-53.

16. Anderson RM, Novacek G, Richards F. Influencing the personal meaning of diabetes: research and practice. Diabetes Educ 1988;14:297-302.

17. Tabak ER. The relationship of information exchange during medical visits to patient satisfaction: a review. Diabetes Educ, 1988;13:36-40.

18. Theman J. Patientutbildning—vad kan sjukvården lära fraan pedagogiken [Patient education—what can be learnt from education science]. Diabetes, Svensk Medicin 1989, Nr 4. Spri, Örebro 1989.

19. Knowles M. The adult learner—a neglected species. New York: Gulf Publishing Corporation, 1978.

20. Gibbins RL, Saunders J. How to do it. Develop diabetic care in general practice. Br Med J 1988;297:187-89.

21. Weinberger M, Cohen SJ, Mazzuca S. The role of physician's knowledge and attitudes in effective diabetes management. Soc Sci Med 1984;19:965-69.

22. Kinmonth A, Marteau TM. Doctors' beliefs and the management of insulin-dependent-diabetes: implications for sharing care. Fam Pract 1989;6:193-98.

23. Reichard P, Britz A, Cars I, Nilsson BY, Sobicinsky-Olsson B, Rosenqvist U. The Stockholm Diabetes Intervention Study (SDIS) 18 months' results. Acta Med Scand 1988;224:115-22.

24. Anderson RM. Development of diabetes attitude scale for health-care professionals. Diabetes Care 1988;12:120-27.

25. Pendleton D. Schofield T, Marinker M, eds. In pursuit of quality. Approaches to performance review in general practice. Exeter, England: The Royal College of General Practitioners, 1986.

26. Lohr NK, Schroeder SA. A strategy for quality assurance in Medicare. N Engl J Med 1990;322:707-12.

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